

**INTRODUCTION TO LIBRARY & INFORMATION SCIENCES (5501)
WORKSHOP SPRING 2013**

COMPUTER HARDWARE & SOFTWARE

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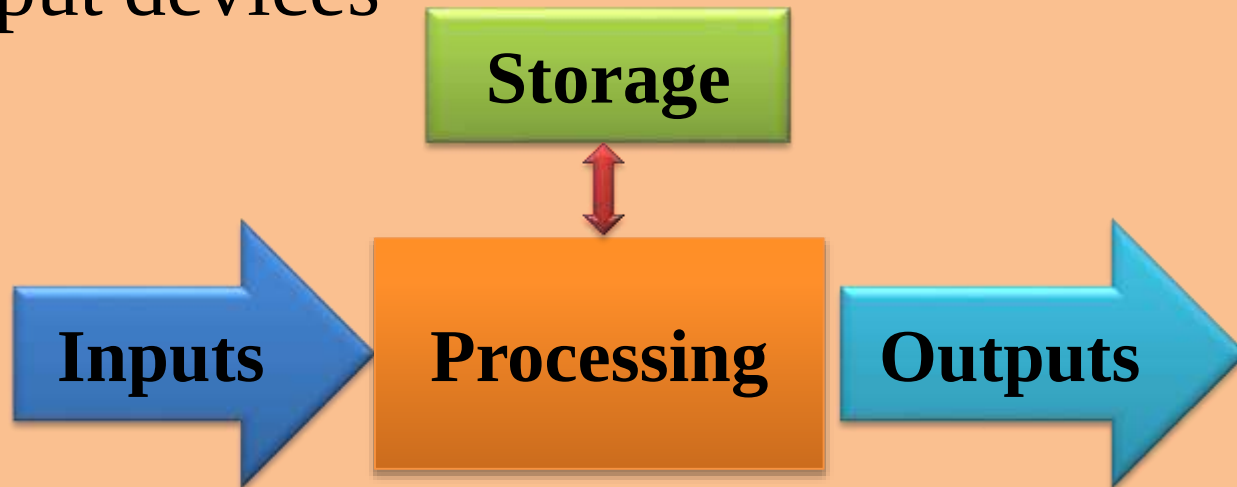
What is COMPUTER?

- Computers are automatic, electronic machines that
 - accept data & instructions from a user (INPUT)
 - store the data & instructions (STORAGE)
 - manipulate the data according to the instructions (PROCESSING)
 - store &/or output the results to the user (OUTPUT)
- A computer system is composed of hardware and software.

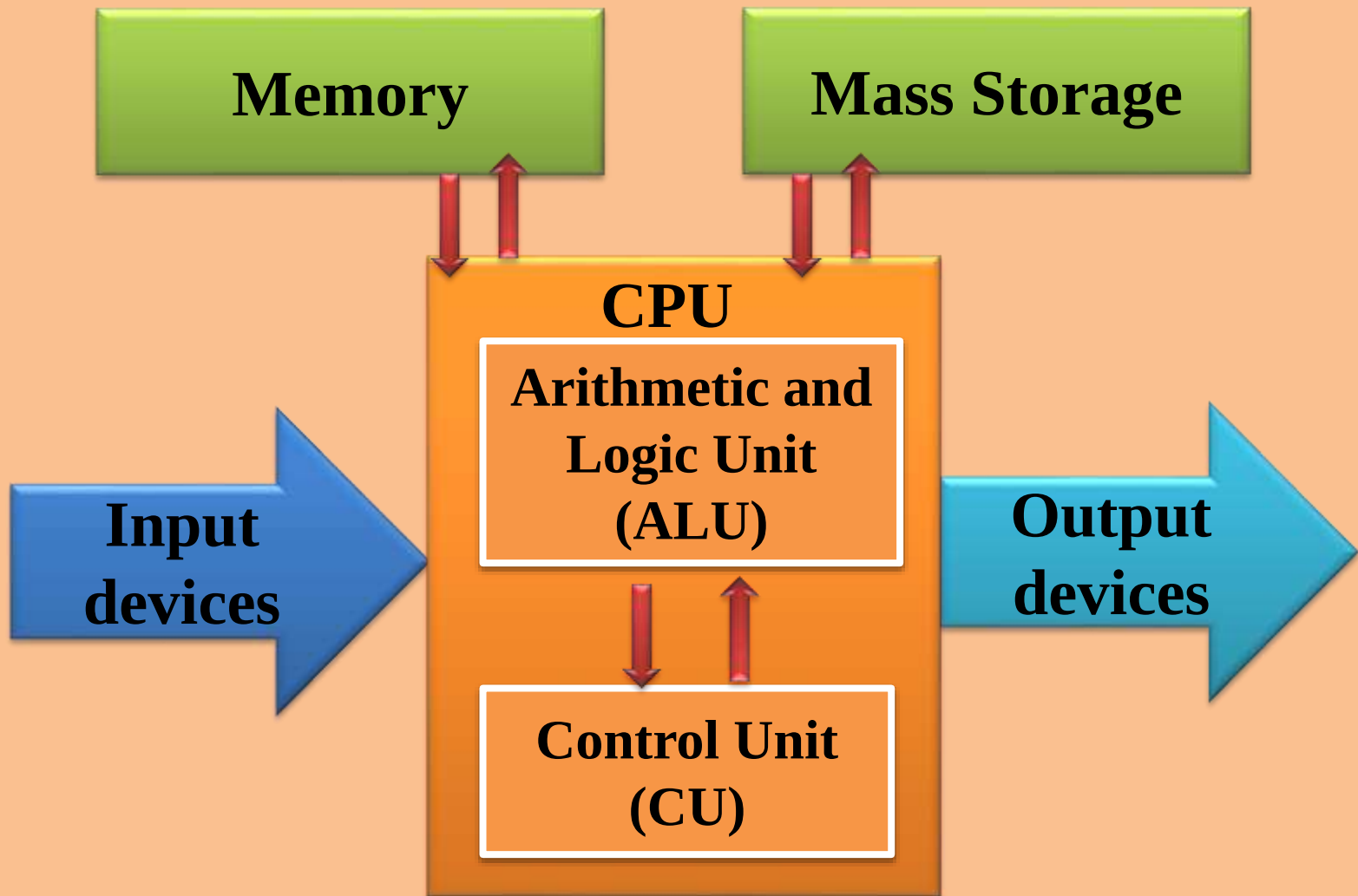
Components of Computer

➤ Its components include...

1. Input devices
2. Central Processing Unit (CPU)
3. Primary storage (Memory)
4. Secondary storage (Mass storage)
5. Output devices



Components of Computer



HARDWARE

- Mechanical, electrical, electronic, or other physical equipment and machinery associated with a computer system or necessary for the playback or projection of nonprintmedia.
- Basic microcomputer hardware includes a central processing unit (CPU), keyboard, and monitor.

INPUT DEVICES

- Input devices allow the user to enter information into the system, or control its operation. Examples ...

Keyboard Mouse Optical Mouse Trackball Joystick



Light Pen Webcam Barcode Reader Touchscreen



OUTPUT DEVICES

- An output device is a piece of hardware that is used to display or output data which has been processed or has been stored on the computer.
Examples are...

Monitor



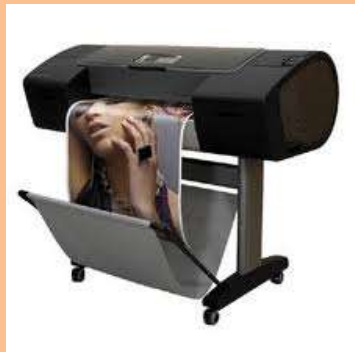
Printer



Speakers



Plotter



Video Beam



CENTRAL PROCESSING UNIT (CPU)

- **CPU is the heart and brain of a computer. It is normally an Intel Pentium (or equivalent).**
- **It receives** data as input.
- **Follows instruction**, and **processes** data accordingly.
- **Presents** the information as out put to the user.
- Its primary function is to **execute** programs.
- It **controls & coordinates** all components such as memory, input and output devices.

- **CPU consists of two units**
 - Control Unit (CU)
 - Arithmetic & Logic Unit (ALU)



Contd... CPU

Control Unit (CU)

The control unit controls the overall activities of the components of the computer. It is mainly used to coordinate the activities among other units. It will send commands signals and controls the sequence of instructions to be executed.

Arithmetic and Logic Unit (ALU)

Performs arithmetic and logical operations on the data.

- arithmetic operations (+, -, *, /)
- logic operations (>, <, <=, >=, etc.)

PRIMARY STORAGE (MEMORY)

- Primary Storage also called **Main Memory**
 - On board memory (located on the motherboard)
 - Very fast, but expensive
 - Two types
 - RAM – Random Access Memory
 - ROM – Read Only Memory
- **RAM - Random Access Memory**
 - Read/write capability
 - Contents lost when computer is turned off (volatile)
 - A program must be in RAM for it to execute
 - Its types are Static & Dynamic RAM

PRIMARY STORAGE (MEMORY)

➤ **ROM - Read Only Memory**

- Read but not write capability
- Permanent (non volatile)
- Stores the preliminary instructions to be executed when the computer is turned on, e.g.
 - To check RAM
 - To check communications with peripheral devices
 - Bootstrap loader program
- Its types are ROM, PROM, EPROM & EEPROM

SECONDARY STORAGE (MASS STORAGE)

- Secondary Storage also **called Mass Storage**
 - External devices (not on the motherboard); either inside or outside the computer
 - Store programs and data permanently
 - Slower, but cheaper

SECONDARY STORAGE (MASS STORAGE)

–Different sizes/styles

- Floppy Disk - 1.4MB (portable)
- Zip Drive - 100-750MB (portable)
- CD - 650MB (portable)
- Hard Disk Drive ≥ 20 GB (not portable)
- Tape - 50GB (portable, very slow)
- Flash drives (portable)

SOFTWARE

- Programs written in a special language with a **series of instructions** to a computer or its peripherals that cause the computer **to solve a problem or perform a task** to achieve a specific set of results.
- Simply a **set of instruction** necessary for a computer to accomplish required tasks.

SOFTWARE

- A computer program is a series of instructions
 - each instruction is expressed in a format consistent with a predefined set of rules
 - a computer processes data under the direction of the instructions in a program
 - there are instructions to input, process, store and output data
 - the user of a program (as distinct from its creator) has no need to be aware of the details of its construction
 - the user is only interested in the services that the program is able to provide

TYPES OF SOFTWARE

➤ **Operating System**

- controls all machine activities
- provides the user interface to the computer
- manages resources such as the CPU and memory
- DOS, Windows XP, Unix, Linux, Mac OS

➤ **Application program**

- generic term for any other kind of software (custom made)
- MS Word, MS Excel, Games,..... Library softwares e.g, Koha, WINISIS, LIMS, LAMP....

Conclusion

- ✓ Computer **hardware** refers to the physical parts or components of a computer.
- ✓ In contrast, **software** is untouchable. Software exists as ideas, application, concepts, and symbols, but it has no substance.
- ✓ A **combination** of hardware and software forms a usable **computing system**.

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THANK YOU



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